LUISIANA INFA-TRACK

DATABASE ARCHITECTURE & SCHEMA SPECIFICATION

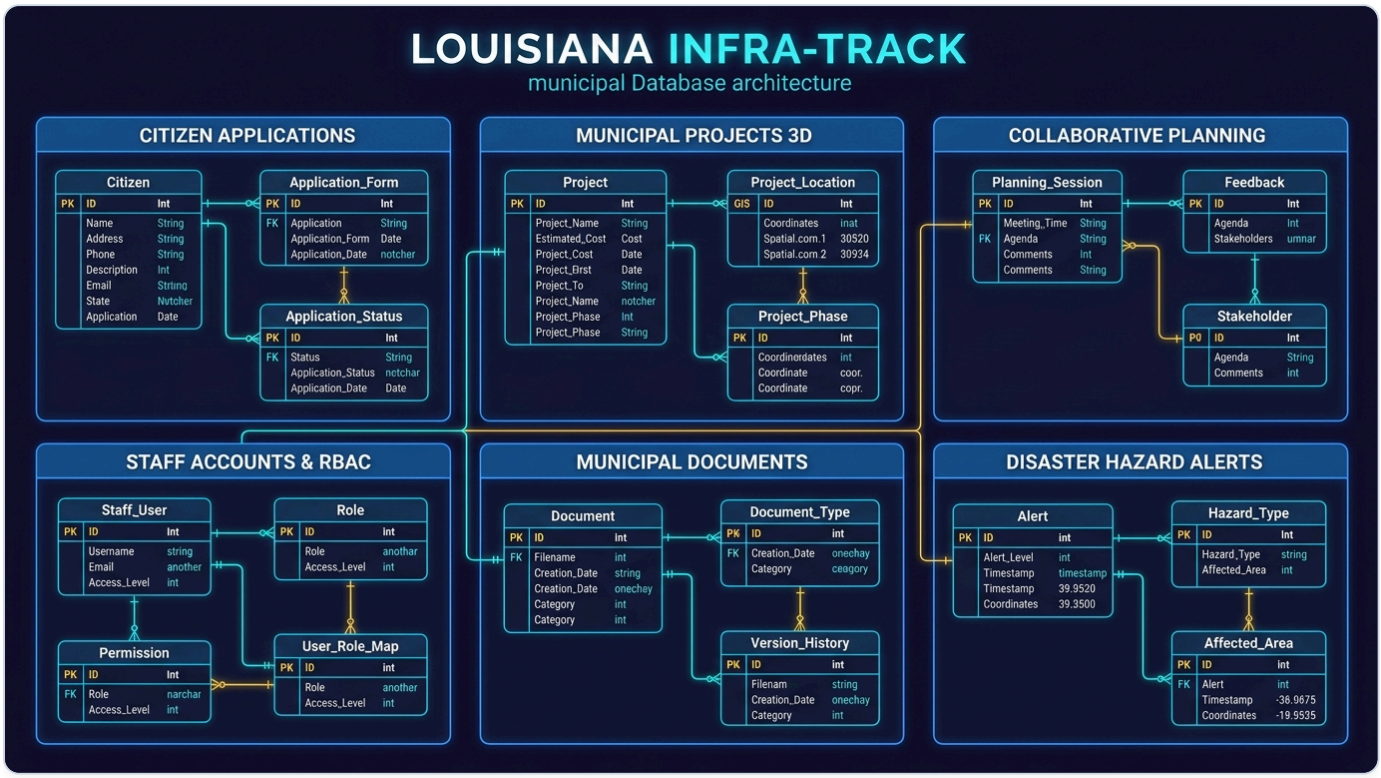
Municipality of Luisiana, Laguna
PSGC: 0403412000 | Region IV-A (CALABARZON)
Target: PostgreSQL 14+ Relational & Spatial

Municipal Enterprise Database Architecture

Unified relational database design powering real-time GIS infrastructure tracking, 3D WebGL model transforms, Citizen's Charter zoning certificates, inter-office collaborative planning (MPDC, Engineering, Agriculture, Treasury), and multi-layered security audit trails.

PostgreSQL 14+ / 16+ | PostGIS Geospatial Ready | 18 Core Tables | 23 Barangays (PSGC) | Cryptographic Data Hash | RBAC Access Control

1. VISUAL DATABASE ARCHITECTURE DIAGRAM



2. SYSTEM CONFIGURATION & GEOGRAPHIC MASTERLISTS

barangays

Official 23 Barangays of Luisiana, Laguna

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
PK id	VARCHAR(64)	PRIMARY KEY	Unique identifier (e.g. 'brgy-1')
code	VARCHAR(32)	UNIQUE, NOT NULL	Barangay code (e.g. 'ZONE-1', 'DELAPAZ')
name	VARCHAR(128)	NOT NULL	Full official name (e.g. 'Barangay Zone I (Poblacion)')
zone	VARCHAR(64)	DEFAULT 'Rural Zone'	Zoning classification ('Urban / Commercial Zone', 'Rural')
is_active	BOOLEAN	DEFAULT TRUE	Active status in drop-downs and GIS filters
created_at / updated_at	TIMESTAMPTZ	DEFAULT NOW()	Auto-managed timestamp

system_config

System maintenance mode, parameters, and metadata

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
PK id	VARCHAR(64)	PRIMARY KEY	Default 'master_config'
version	VARCHAR(32)	NOT NULL	Software version release (e.g. '1.2.0')
maintenance_enabled	BOOLEAN	DEFAULT FALSE	Maintenance barrier toggle for public portal
maintenance_message	TEXT	NULL	Notice message displayed to public users
config_json	JSONB	DEFAULT '{}'	Extended key-value system configuration
updated_at	TIMESTAMPTZ	DEFAULT NOW()	Timestamp of last modification

system_masterlists

Zoning types, building classifications, and funding sources

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
PK id	VARCHAR(64)	PRIMARY KEY	Unique item ID (e.g. 'zc-1', 'bt-1')
category	VARCHAR(64)	NOT NULL	Category: buildingTypes, zoningClassifications, permitCategories
code	VARCHAR(64)	NOT NULL	Unique slug within category (e.g. 'URBAN_COMM')
name	VARCHAR(255)	NOT NULL	Human-readable label
zone_group	VARCHAR(64)	NULL	High-level group (Commercial, Residential, Agricultural)
is_active	BOOLEAN	DEFAULT TRUE	Whether selectable in application intake

3. AUTHENTICATION, RBAC & SECURITY AUDIT LOGS

staff_accounts

LGU Staff & Official user profiles (PBKDF2 secured)

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
PK id	UUID	PRIMARY KEY	Default uuid_generate_v4()
UK username	VARCHAR(64)	UNIQUE, NOT NULL	Login username ('mpdc', 'engineer', 'agriculture', etc.)
name	VARCHAR(255)	NOT NULL	Full name / designation of employee
employee_id	VARCHAR(64)	NULL	LGU Employee ID (e.g. 'MPDC-001')
role	VARCHAR(64)	CHECK (role IN ...)	RBAC role (Super Admin, MPDC, Engineer, Treasury, Assessor)
department	VARCHAR(255)	NOT NULL	Department or municipal office name
password_hash	TEXT	NOT NULL	PBKDF2-SHA256 password hash
password_salt	TEXT	NOT NULL	Cryptographic salt (base64)
iterations	INT	DEFAULT 120000	PBKDF2 iteration count
failed_attempts	INT	DEFAULT 0	Failed login counter for brute-force lockout
locked_until	TIMESTAMPTZ	NULL	Lockout expiration timestamp
must_change_password	BOOLEAN	DEFAULT FALSE	Enforces password change upon initial login
is_active	BOOLEAN	DEFAULT TRUE	Account enablement toggle

user_sessions

Active staff sessions with network IP and device fingerprinting

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
PK id	VARCHAR(128)	PRIMARY KEY	Session ID / Bearer token
FK username	VARCHAR(64)	REFERENCES staff_accounts	User associated with session
ip_address	VARCHAR(64)	NULL	Client IP address (resolved to LGU Intranet)
device_name	VARCHAR(128)	NULL	Friendly device name (e.g. 'Windows PC (Chrome)')
device_type	VARCHAR(32)	NULL	Form factor: desktop, mobile, tablet
location_label	VARCHAR(255)	NULL	Resolved physical network (e.g. 'Munisipyo ng Luisiana')
expires_at	TIMESTAMPTZ	NOT NULL	Session expiration timestamp (8-hour TTL)

audit_logs & security_quarantine_logs

System activity audit trail & anti-malware quarantine

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
PK id	VARCHAR(64)	PRIMARY KEY	Log event ID
actor_username	VARCHAR(64)	NULL	User or service responsible for action
action_type	VARCHAR(64)	NOT NULL	Action type (LOGIN, CREATE, UPDATE, APPROVE, QUARANTINE)
resource_type	VARCHAR(64)	NOT NULL	Entity (APPLICATION, PROJECT, PROPOSAL, AUTH)
details	JSONB	DEFAULT '{}'	Structured payload diff / inspection details
threat_type	VARCHAR(128)	quarantine only	PDF_JAVASCRIPT, POLYGLOT, EXECUTABLE_ATTACHMENT
created_at	TIMESTAMPTZ	DEFAULT NOW()	Timestamp of event

4. CITIZEN APPLICATIONS & PERMITTING (CITIZEN'S CHARTER)

citizen_applications — Online zoning, land titling, and research permit applications

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
PK id	VARCHAR(64)	PRIMARY KEY	Unique system application ID
UK tracking_number	VARCHAR(64)	UNIQUE, NOT NULL	Public tracking number (e.g. 'LUIS-ZC-2026-9559')
service_type	VARCHAR(64)	NOT NULL	zoning_certificate, land_titling, planning_research
category	VARCHAR(64)	NOT NULL	private_infrastructure, agricultural, municipal_project
status	VARCHAR(64)	CHECK (status IN ...)	submitted, in_review, ocular_inspection, approved, etc.
applicant_name	VARCHAR(255)	NULL	Proponent / property owner name
contact_phone / email	VARCHAR(128)	NULL	Contact details for SMS / email notifications
barangay	VARCHAR(128)	NULL	Barangay where property is located
latitude / longitude	DOUBLE PRECISION	NULL	WGS84 GIS coordinates for spatial pinpointing
estimated_cost	NUMERIC(15,2)	DEFAULT 0	Estimated project / construction cost (PHP)
lot_details	JSONB	DEFAULT '{}'	TCT No., Tax Dec No., lot area sq.m., building type
step_progress	JSONB	DEFAULT '{}'	Citizen charter 5-step milestone tracking
engineer_approved	BOOLEAN	DEFAULT FALSE	Municipal Engineer verification flag for map pinning
payment_status	VARCHAR(64)	DEFAULT 'unpaid'	unpaid, order_of_payment_issued, paid
or_number	VARCHAR(128)	NULL	Official Receipt (O.R.) number from Municipal Treasury
data_hash	VARCHAR(64)	NULL	SHA-256 tamper-evident digital fingerprint

site_inspection_photos & application_remarks — 360° Street panoramas, side inspection photos, and staff remarks

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
PK id	VARCHAR(64)	PRIMARY KEY	Unique record ID
FK application_id	VARCHAR(64)	REFERENCES citizen_applications	Parent application foreign key (CASCADE delete)
tracking_number	VARCHAR(64)	NULL	Denormalized tracking number for indexed queries
photo_url	TEXT	photo table	File URL or storage path of site panorama
photo_type	VARCHAR(50)	photo table	'street_panorama' (360 view) or 'side_photo'
stage	VARCHAR(50)	photo table	Inspection stage: 'before', 'during', 'after'
progress	INT	photo table	Observed construction completion percentage (0-100)
from_office	VARCHAR(64)	remarks table	Office authoring remark: MPDC, Engineer, Treasury
message	TEXT	remarks table	Official message, deficiency note, or feedback

5. MUNICIPAL INFRASTRUCTURE PROJECTS & 3D WEBGL VISUALIZATION

municipal_projects — Infrastructure, buildings, and spatial assets rendered on the 3D map

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
PK id	VARCHAR(64)	PRIMARY KEY	Project ID (e.g. 'PRJ-101', 'app-179007...')
name	VARCHAR(255)	NOT NULL	Project title (e.g. 'Luisiana RHU Expansion')
model_type	VARCHAR(64)	NOT NULL	3D Model: municipal_hall, rhu, school, road, bridge, barn
type	VARCHAR(64)	DEFAULT 'Municipal Project'	Classification: Municipal Project, Private Building, Agri
department	VARCHAR(64)	NOT NULL	Responsible department: MPDC, Engineering, Agriculture
status	VARCHAR(64)	CHECK (status IN ...)	Planned, Ongoing, Delayed, Completed, Suspended
progress	INT	0 - 100	Overall construction / execution progress %
latitude / longitude	DOUBLE PRECISION	NOT NULL	WGS84 coordinates of 3D asset anchor point
rotation_yaw/pitch/roll	DOUBLE PRECISION	DEFAULT 0	3D spatial rotations in degrees for Three.js alignment
model_scale_x/y/z	DOUBLE PRECISION	DEFAULT 1	World-space dimensions & scale multipliers
model_height	DOUBLE PRECISION	DEFAULT 0	Height offset above terrain in meters (0 = ground clamped)
map_shape / map_sketch	JSONB	NULL	Procedural shape footprints (polygons, lines, pins)
budget_total / spent	NUMERIC(15,2)	NULL	Allocated budget vs actual spent (PHP)
contractor / funding	VARCHAR(255)	NULL	Awarded contractor & funding source (20% DF, GenFund)

project_milestones, project_issues, & project_photos — Milestone schedules, contractor delays, and drone progress photos

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
PK id	VARCHAR(64)	PRIMARY KEY	Record ID
FK project_id	VARCHAR(64)	REFERENCES municipal_projects	Parent project foreign key (CASCADE delete)
title	VARCHAR(255)	milestones / issues	Milestone name or bottleneck description
target_date / completed_at	DATE / TIMESTAMPTZ	milestones	Target delivery date & actual completion timestamp
kind	VARCHAR(32)	issues / photos	'delay' / 'issue' OR photo kind: 'site' / 'progress'
photo_url	TEXT	photos	Image asset storage path
latitude / longitude	DOUBLE PRECISION	photos	Inspector GPS position when photo was captured

6. COLLABORATIVE PLANNING, DMS & CITIZEN ENGAGEMENT

planning_proposals, comments, & approvals — Inter-office project proposals, committee reviews, and voting

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
PK id	VARCHAR(64)	PRIMARY KEY	Proposal ID (e.g. 'prop-101')
title	VARCHAR(255)	NOT NULL	Proposal title
priority	INT	CHECK (1 - 5)	Priority rating: 1 (Lowest) to 5 (Critical Emergency)
status	VARCHAR(64)	CHECK (status IN ...)	draft, submitted, in_review, recommended, approved, rejected
request_kind	VARCHAR(64)	DEFAULT 'office_proposal'	barangay_request, office_proposal, zoning_certificate
FK submitter_username	VARCHAR(64)	NOT NULL	Author staff account username
needs_assessment	JSONB	DEFAULT '{}'	Population served, hazard exposure, urgency notes
recommendation_score	DOUBLE PRECISION	NULL	AI / committee calculated multi-criteria score

municipal_documents & citizen_engagements — Official DMS records (CLUP, AIP) and public feedback

COLUMN	DATA TYPE	CONSTRAINTS	DESCRIPTION
PK id	VARCHAR(64)	PRIMARY KEY	Document or Engagement submission ID
category	VARCHAR(64)	NOT NULL	CLUP, CDP, LDIP, AIP, engineering_drawing, permit
file_url / file_name	TEXT / VARCHAR	documents	Secure file download path and filename
version	INT	DEFAULT 1	Revision version number with version history table
kind	VARCHAR(32)	engagements	Public submission kind: 'feedback', 'issue', 'suggestion'
status	VARCHAR(32)	engagements	Review state: 'new', 'reviewing', 'resolved', 'dismissed'
staff_note	TEXT	engagements	Official internal LGU response notes